

Lineare Algebra
Serie 25
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1.

Assume U is invariant under T .

$$\forall v \in U, \quad \forall w \in U^\perp$$

$$0 = \langle v, w \rangle = \langle Tv, w \rangle = \langle v, T^*w \rangle$$

Since v, w are arbitrary and $\langle v, T^*w \rangle = 0 \Rightarrow U^\perp$ is invariant under T^* .

For the other direction we can do exactly the same proof.

□

2.

define

$$t = f \circ g_1 - f \circ g_2$$

From the exercise we know

$$\begin{aligned} f^* \circ f \circ g_1 &= f^* \circ f \circ g_1 \\ \Rightarrow f^* \circ t &= 0 \end{aligned}$$

$$\forall v \in V$$

$$0 = \langle f^*(t(v)), v \rangle = \langle h(v), f(v) \rangle$$

set

$$u = g_1(v) - g_2(v)$$

since f is linear

$$\begin{aligned} t(v) &= f \circ g_1(v) - f \circ g_2(v) \\ &= f \circ (g_1(v) - g_2(v)) = f(u) \end{aligned}$$

we get

$$\begin{aligned} 0 &= \langle f^*(t(v)), u \rangle = \langle h(v), f(u) \rangle = \langle h(v), h(v) \rangle \\ &\Rightarrow \|h(v)\| = 0 \\ &\Rightarrow h(v) = 0 \\ &\Rightarrow f \circ g_1 = f \circ g_2 \end{aligned}$$

3.

since $T \in \text{End}(V)$ a normal operator.

$$T^*T = TT^*$$

$\Rightarrow$ there exists an orthonormal basis made with eigenvectors

Additionally, we know that V is equal to a finitely many orthogonal subspaces. Define W_i equal to the span of the i -th orthonormal basis vector.

$$W_i = \text{span}(e_i)$$

$$V = W_1 \oplus \dots \oplus W_k$$

Every vector $v \in V$ can be written with a sum

$$v = \sum_{i=1}^n w_i, \quad w_i \in W_i$$

With

$$P_{W_i}(v) = w_i$$

We have

$$T(w_i) = \lambda_i w_i, \quad w_i \in W_i$$

$$\forall v \in V$$

$$\begin{aligned} \Rightarrow T(v) &= \sum_{i=1}^k T(w_i) \\ &= \sum_{i=1}^k \lambda_i w_i \\ &= \sum_{i=1}^k \lambda_i P_{W_i}(v) \end{aligned}$$

4.

4.a.

$$T \text{ is self adjoint} \Leftrightarrow T = T^*$$

$$\text{let } p_1(x) = 1, p_2(x) = x$$

$$\langle Tp_1, p_2 \rangle = \int_0^1 0 dx = 0$$

$$\langle p_1, T^*p_2 \rangle = \langle p_1, Tp_2 \rangle = \int_0^1 x dx = \frac{1}{2}$$

Contradiction $\Rightarrow T$ is not self adjoint

4.b.

$(1, x, x^2)$ is not an orthonormal basis.

$$[T^*] = [T]^*$$

is only true if T is a Matrix with respect to an orthonormal basis

5.

5.a.

Since $\dim(V) < \infty$ there exists a basis $\mathcal{B}$ for V

$$\mathcal{B} = \{e_1, \dots, e_n\}$$

Define φ . Let $w \in W, v \in V$

$$\begin{aligned} \varphi_w(v) : V &\rightarrow \mathbb{R} \\ v &\mapsto \langle f(v), w \rangle \end{aligned}$$

A linear map

We want to find the unique adjoint map of f , which must fulfill:

$$\varphi_w(v) = \langle f(v), w \rangle = \langle v, f^*(w) \rangle$$

Choose an orthonormal basis $\mathcal{B} = \{e_1, \dots, e_n\}$ for V . Then $\forall v \in V$

$$v = \sum_{i=1}^n \lambda_i e_i$$

Since φ_w is linear

$$\varphi_w(v) = \sum_{i=1}^n \lambda_i \varphi_w(e_i)$$

define u

$$u = \sum_{i=1}^n \varphi_w(e_i) e_i$$

Since the basis $\mathcal{B}$ is orthonormal, most terms vanish.

$$\langle v, u \rangle = \sum_{i=1}^n \sum_{k=1}^n \varphi_w(v) \lambda_i \langle e_i, e_k \rangle = \sum_{i=1}^n \sum_{k=1}^n \lambda_i \varphi_w(e_k) \delta_{i,k} = \sum_{i=1}^n \lambda_i \varphi_w(e_i) = \varphi_w(v)$$

Therefore, exists such a u .

We want to show the uniqueness of this vector $u \in V$.

Assume we have two vectors u_1, u_2 such that

$$\begin{aligned} \langle v, u_1 \rangle &= \langle v, u_2 \rangle \\ \Rightarrow 0 &= \langle v, u_1 - u_2 \rangle \end{aligned}$$

This must be true for all $v \in V$.

Set

$$\begin{aligned} v &= u_1 - u_2 \\ \Rightarrow \langle v, v \rangle &= 0 \\ \Rightarrow \|u_1 - u_2\|^2 &= 0 \\ u_1 &= u_2 \end{aligned}$$

$\Rightarrow \exists! f'$

5.b.

6.

6.a.

$$D(f) = \frac{df}{dx}$$

using integration by parts

$$\begin{aligned} \langle D(f(x)), g(x) \rangle &= \frac{1}{2\pi} \int_0^{2\pi} f'(x) g(x) dx \\ &= \frac{1}{2\pi} (f(x) g(x)) \Big|_0^{2\pi} - \int_0^{2\pi} f(x) g'(x) dx \\ &= 0 - \frac{1}{2\pi} \int_0^{2\pi} f(x) g'(x) dx \\ &= \langle f(x), -D(g(x)) \rangle \end{aligned}$$

$\Rightarrow D$ is not self adjoint

$$D^*(f) = -D(f)$$

6.b.

$$\begin{aligned} \langle -\Delta(f), g \rangle &= \langle D(f), D(g) \rangle \\ &= \langle f, -D \circ D(g) \rangle \\ &= \langle f, \Delta(g) \rangle \end{aligned}$$

$\Rightarrow \Delta$ is self-adjoint

6.c.

Claim all functions in U are mutually orthogonal.

1. $n = 0$

$\sin(0) = 0$ can't be an eigenvector

$\cos(0) = 1$

2. $n \neq 0$

$\sin(nx)$

$\cos(nx)$

$$\begin{aligned}\langle 1, \sin(nx) \rangle &= \frac{1}{2\pi} \int_0^{2\pi} \sin(nx) dx \\ &= \frac{1}{2\pi n} - \cos(nx) \Big|_0^{2\pi} = 0\end{aligned}$$

$$\langle 1, \cos(nx) \rangle = \frac{1}{2\pi} \int_0^{2\pi} \cos(nx) dx = 0$$

$$\cos(nx) \sin(nx) = \frac{1}{2} (\sin((n+n)x) + \sin((n-n)x))$$

$$\begin{aligned}\Rightarrow \langle \sin(nx), \cos(nx) \rangle &= \frac{1}{2\pi} \int_0^{2\pi} \sin(nx) \cos(nx) dx \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} \sin(2nx) dx = 0\end{aligned}$$

We must norm the functions

$$||\langle 1, 1 \rangle|| = \frac{1}{2\pi} \int_0^{2\pi} 1 \cdot dx = 1$$

using $\sin^2(t) = \frac{1-\cos(2t)}{2}$ and $\cos^2(t) = \frac{1+\cos(2t)}{2}$

$$\begin{aligned}||\sin(nx)||^2 &= \frac{1}{2\pi} \int_0^{2\pi} \sin^2(nx) dx \\ &= \frac{1}{4\pi} \left(\int_0^{2\pi} 1 dx - \int_0^{2\pi} \cos(2nx) dx \right) = \frac{1}{2}\end{aligned}$$

same for $\cos^2(nx)$

$$\begin{aligned}||\cos(nx)||^2 &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1+\cos(2nx)}{2} dx \\ &= \frac{1}{4\pi} \left(\int_0^{2\pi} 1 dx + \int_0^{2\pi} \cos(2nx) dx \right) = \frac{1}{2}\end{aligned}$$

We get and an orthonormal Basis $\mathcal{B}$

$$\mathcal{B} = \{1\} \cup \left\{ \sqrt{2} \cos(nx), \sqrt{2} \sin(nx) \mid n \in \mathbb{N} \setminus \{0\} \right\}$$

Consider the function $\Delta := -D \circ D = -\frac{d^2}{dx^2}$.

First case $e_0 = 1$

$$\Delta e_0 = 0$$

$$\Rightarrow E_0 = \text{span}\{1\} \quad \text{with multiplicity} = 1$$

Case 2 $n \geq 1 \mid n \in \mathbb{N}$

$$\Delta \cos(nx) = n^2 \cos(nx)$$

$$\Delta \sin(nx) = n^2 \sin(nx)$$

Since $\cos$ and $\sin$ are linear independent

$$E_{n^2} = \text{span}\{\sin(nx), \cos(nx)\} \quad \text{with multiplicity} = 2$$

7. Multiple Choice

7.a.

7.a.a. true

7.a.b. false

7.a.c. true

7.b.

7.b.a. false

7.b.b. true

7.b.c. true

7.b.d. true

7.c.

7.c.a. false

7.c.b. true

7.c.c. true

7.c.d. true

7.c.e. true